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SEEK WISDOM, ELEVATE YOUR INTELLECT AND SERVE HUMANITY !



Mandibular and Maxillary Orthognathic Surgery

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Outline:

- Introduction
- Orthognathic surgical procedures of the mandible
- Orthognathic surgical procedures of the maxilla
- Indications
- complications



Mandibular Orthognathic Surgery

Classification

Ramus procedures

- Sagittal split ramus osteotomy
- Vertical subsigmoid osteotomy
- Inverted L-osteotomy
- C- osteotomy



Body proceduers

- Anterior to mental foramen:
 - Step osteotomy/ostectomy
 - Midline symphyseal osteotomy
- Posterior to mental foramen
 - Y ostectomy
 - Rectangular ostectomy
 - Trapezoid ostectomy
 - Inverted V-ostectomy



Sub-apical body procedures

- Anterior
 - Kole osteotomy
 - Combined with midline symphyseal osteotomy
- Posterior
- Total
- Mandibuloplasty

Genioplasty

- Functional genioplasty
- Augmentation genioplasty
- Reduction genioplasty
- Asymmetric genioplasty

Anatomic and Physiologic Considerations of Mandibular Surgeries

Vascular Supply

- blood supply to the body of the mandible under normal conditions comes almost entirely from the inferior alveolar artery.
- Given the available research, it is wise to minimize the periosteal and muscle attachment stripping on the medial surface of the proximal fragment in any of the vertical ramus osteotomies.
- The pulpal blood supply of a tooth should not be affected if a cut is made at least 5 mm away from the apex of the tooth.

Two nerves of concern:

- The marginal mandibular branch of the facial nerve and the inferior alveolar branch of the mandibular nerve.
- The marginal mandibular branch is usually only at risk during extraoral procedures.
- The course of the inferior alveolar nerve into the vertical ramus and then through the body of the mandible makes it extremely susceptible to damage.
- The surgeon's main goal relative to this nerve is to minimize the trauma because its avoidance is almost impossible.
- Exposing the nerve seems to increase the chance for postoperative sensory deficiency.

The Muscle

- Orthognathic surgery affects muscles in primarily two ways: it **changes the length** of a muscle or it **changes the direction** of muscle function.
- Pulling muscles of mastication causes postoperative relapse.
- The chin segment may be displaced by pull from suprahyoid muscles .
- Two techniques have been attempted to improve the stability of mandibular osteotomies in the initial period of greatest amount of relapse (first 3 to 6 weeks after surgery) : -**external supporting mechanisms and internal rigid fixation.**

Osteotomy Techniques

- *Vertical Ramus Osteotomy (subsigmoid)*

Indications:

- Setback procedures

- Mandibular asymmetries

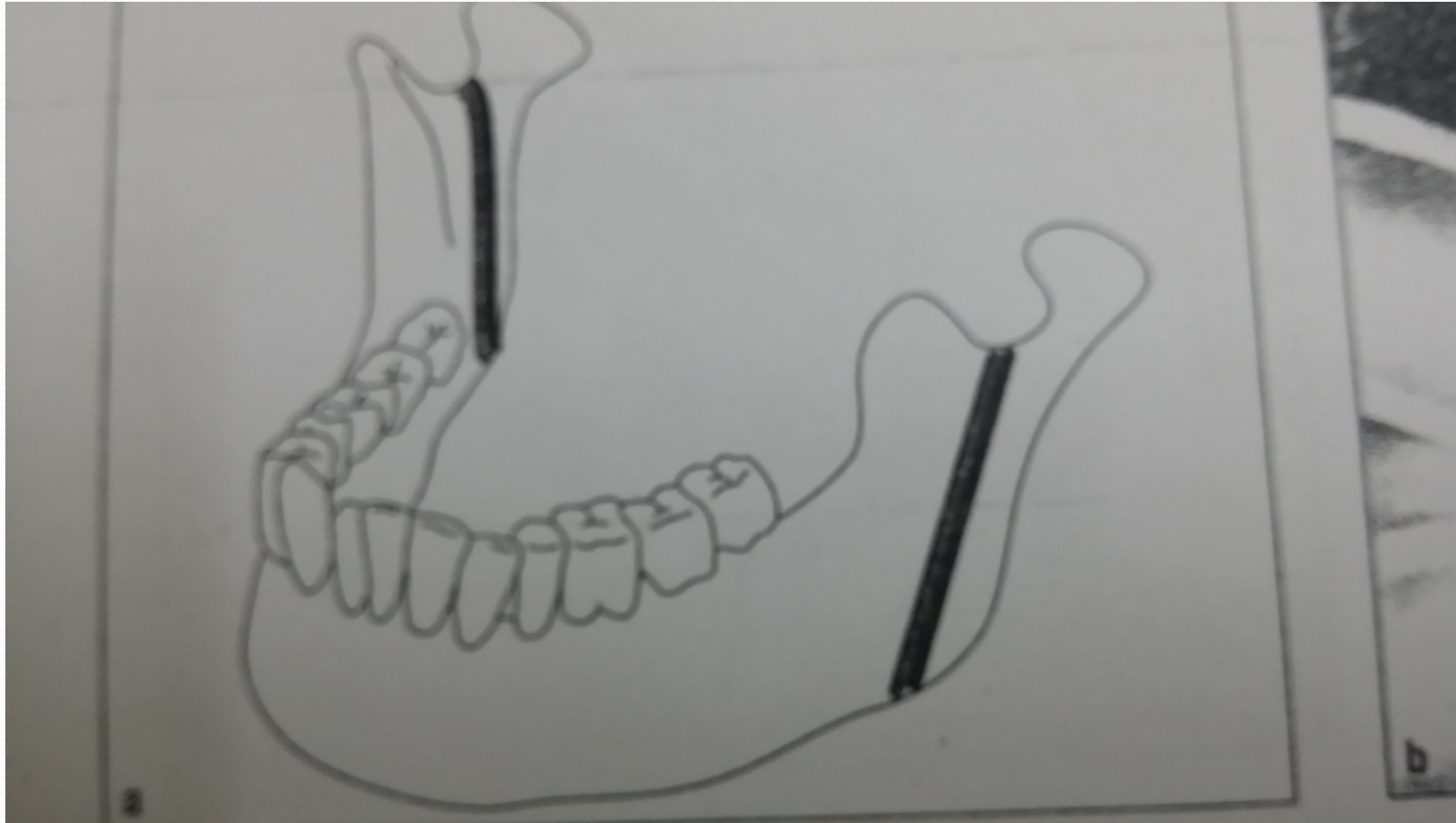
- ❖ It can not be used for mandibular advancement and is not recommended for correction of anterior open bites.
- ❖ Maxillo-mandibular fixation required for six weeks postoperatively

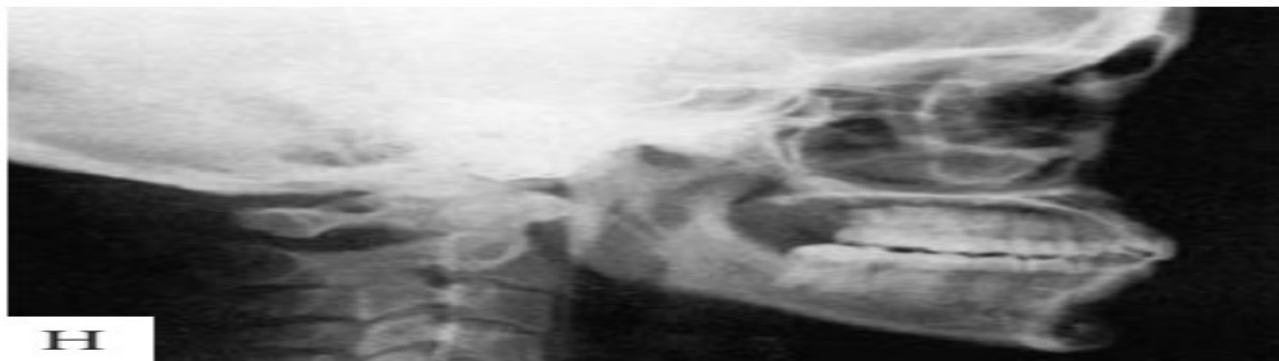
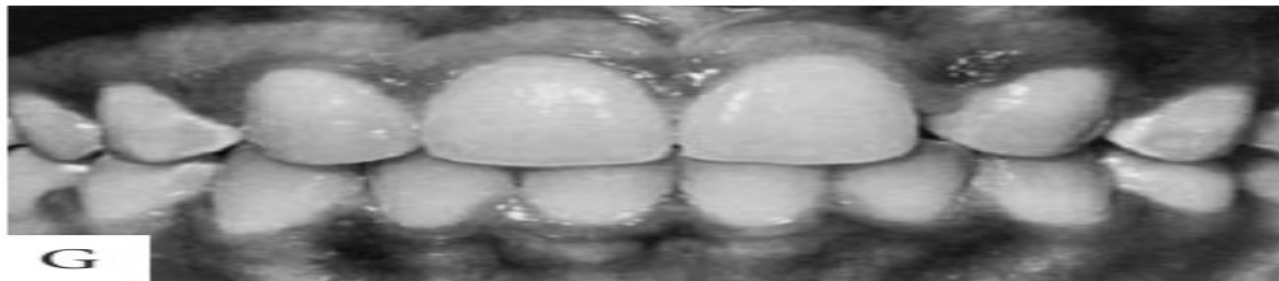
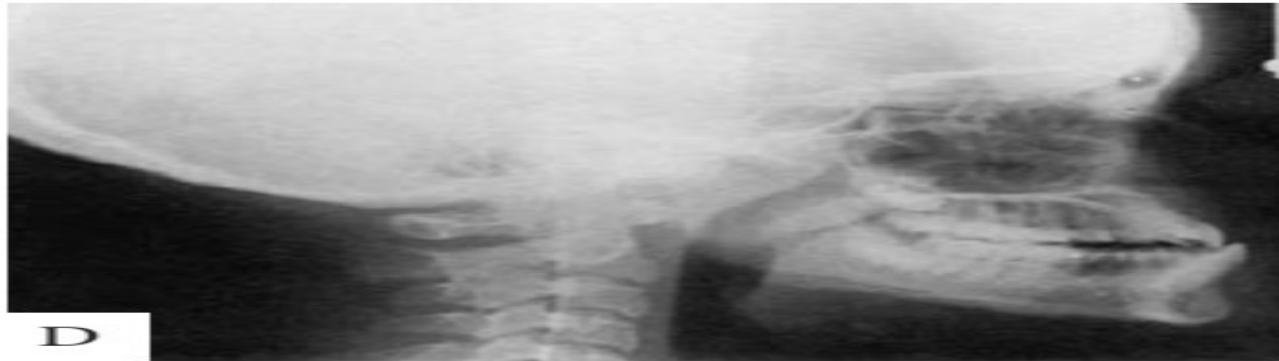
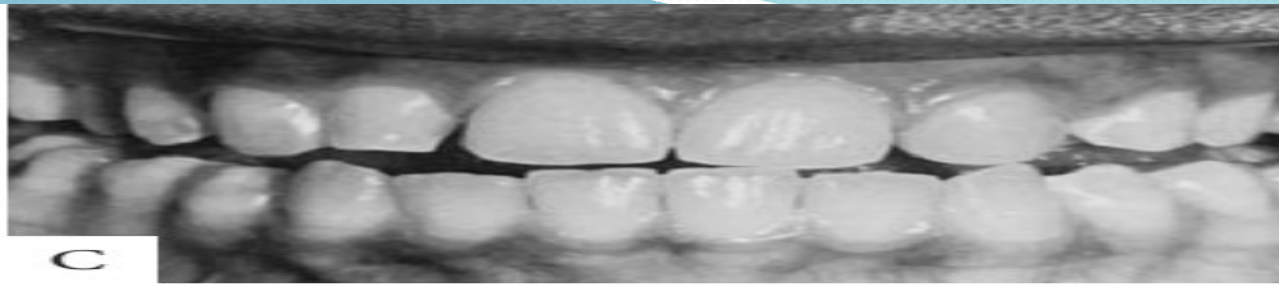
Surgical procedure

- **Extra-oral approach (unusual cases only)**
- Utilize Ridson(submandibular) approach
- Disadvantages:
 - External scars
 - Damage to the marginal mandibular branch of the facial nerve
- **Transoral approach**
- Intra-oral incision extends from the ascending ramus to the first molar
- Avoid extensive periosteum and muscle stripping from ramus.

- Vertical bone cut is done from the sigmoid notch to the angle of the mandible.
- Setback the mandible so that the proximal fragment overlaps the distal fragment on the outer aspect.
- Lateral surface decortication on the distal fragment to establish good bony contact and healing and on the proximal fragment to prevent deformity at the angle.
- Setbacks greater than 1 cm obviates coronoidectomy to prevent excessive tension on the temporalis muscle.
- Establish maxillomandibular fixation for six weeks.

Design of vertical ramus osteotomy





Bilateral Sagittal Split Osteotomy

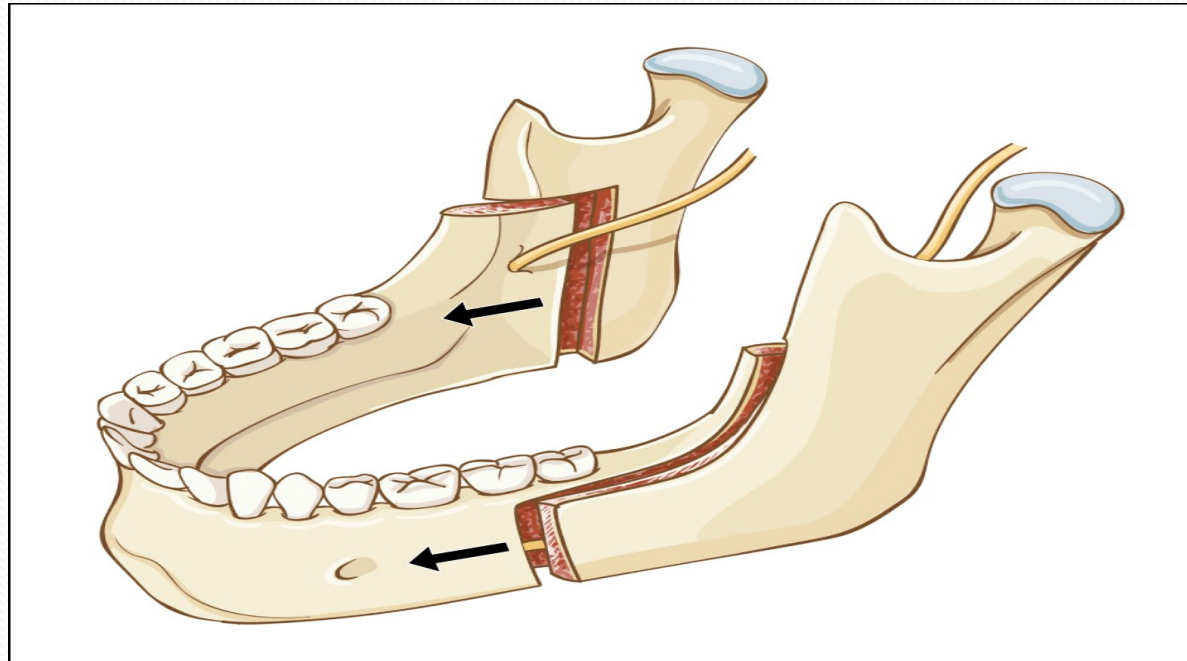
Indications:

- Mandibular setback
- Mandibular advancement
- Mandibular rotation

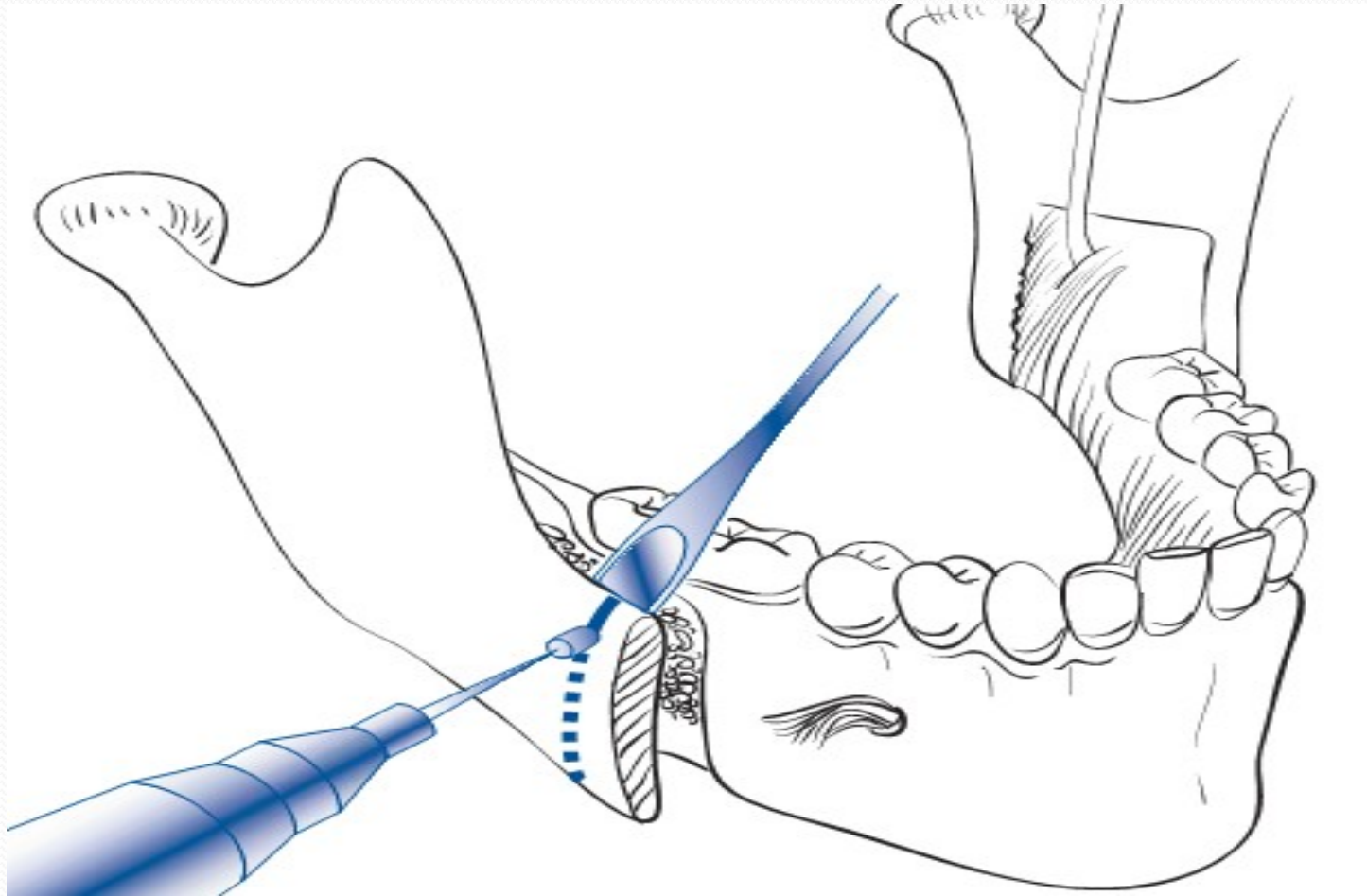
Surgical procedure

- Mucoperiosteal incision commences at the lateral crest of alveolus, extends up the anterior border of the ramus and coronoid process and anteriorly along external oblique ridge as far forward as the first molar.

Bilateral sagittal split osteotomy: lateral and medial bony cuts

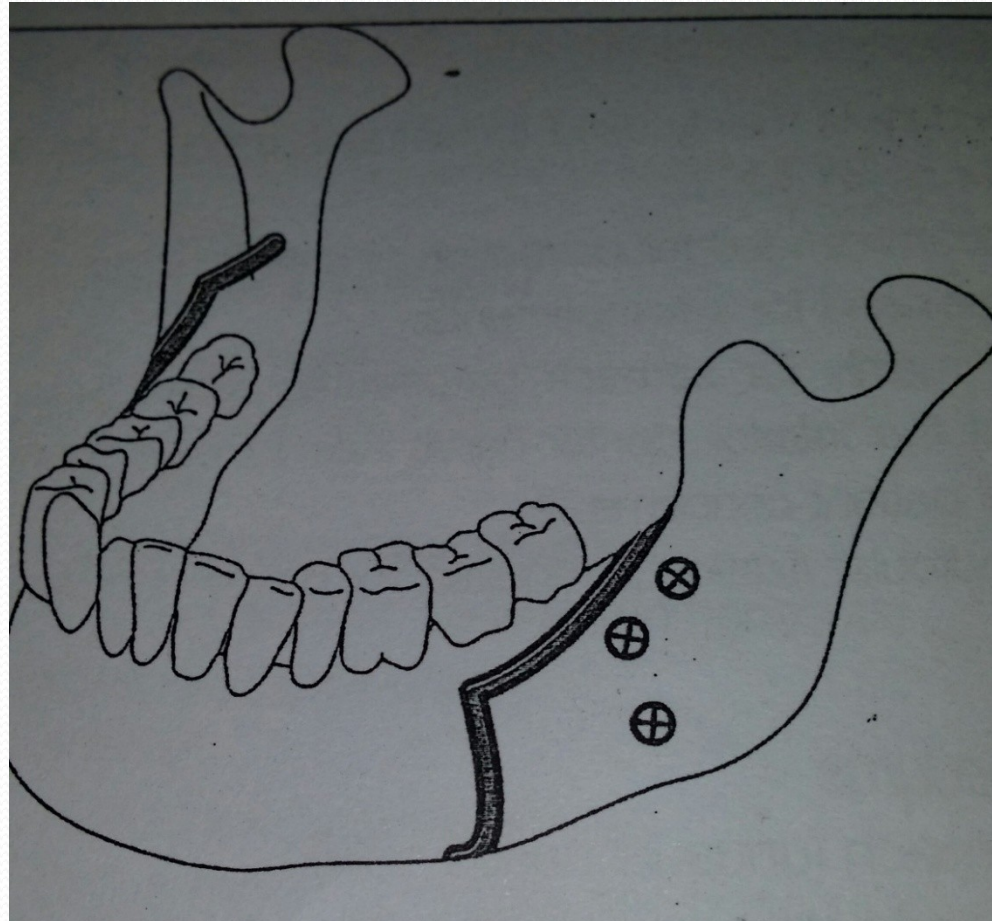


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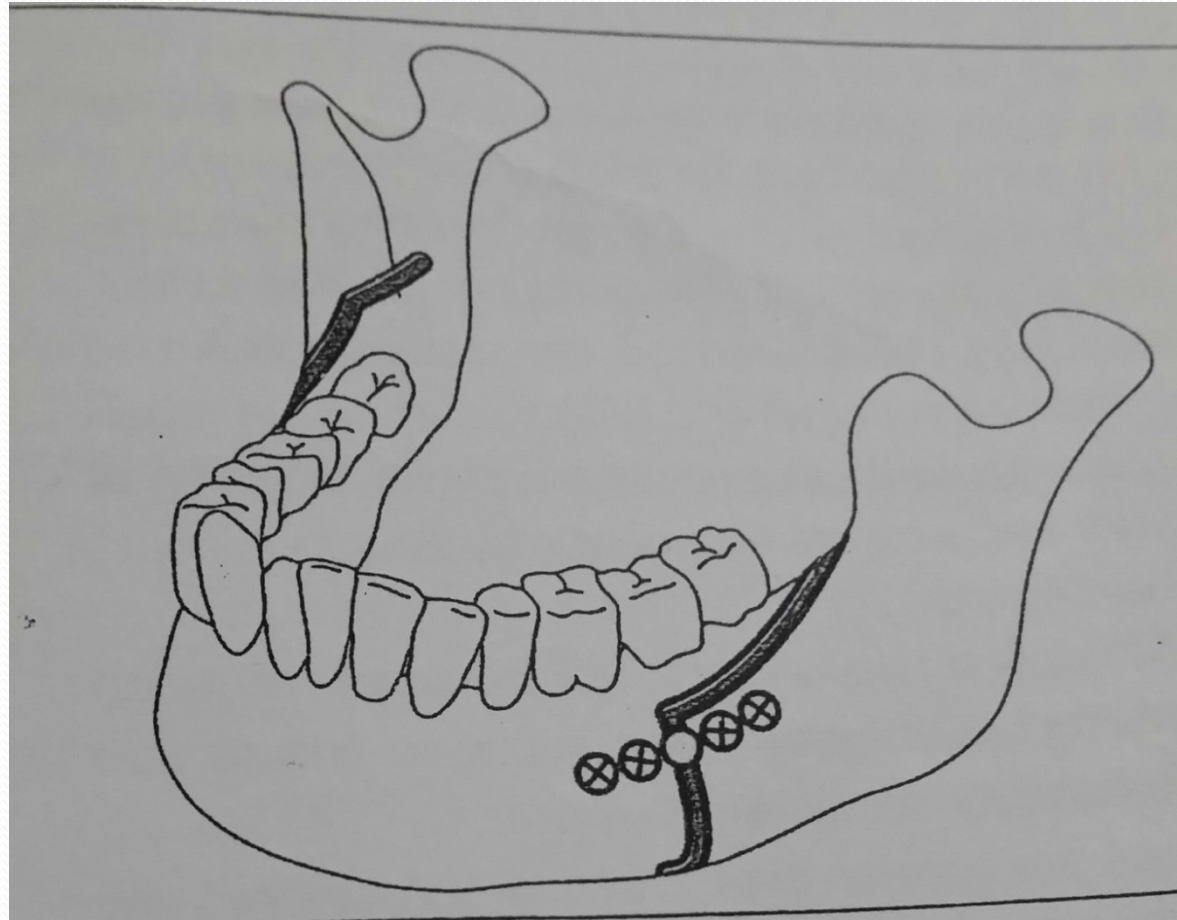


- Lingually make the osteotomy cut from the anterior border of the ramus, backward to just posterior and superior to the lingula.
- Buccally : along the anterior border of the ramus, medial to the external oblique ridge, forward to about the second molar
- Vertically cut the external cortex of the body to and through the lower border of the mandible.
- Begin the split from the lower border of the vertical cut and then along the alveolar cut using fine osteotomes
- The neurovascular bundle must always lie on the medial side in the distal fragment.
- 4-6 weeks maxillomandibular fixation.
- 3 Bicortical screws required (better placed in triangular pattern).
- 1-2 monocortical plates and screws may be used but not as strong as bicortical screws.

Three bicortical screws placed in a triangular pattern



Placement of miniplates and monocortical screw at least 2 monocortical screws should be placed on each side of the osteotomy and the plates positioned on the upper border of the osteotomy site





Inverted L and C Ramus Osteotomies

Indications:

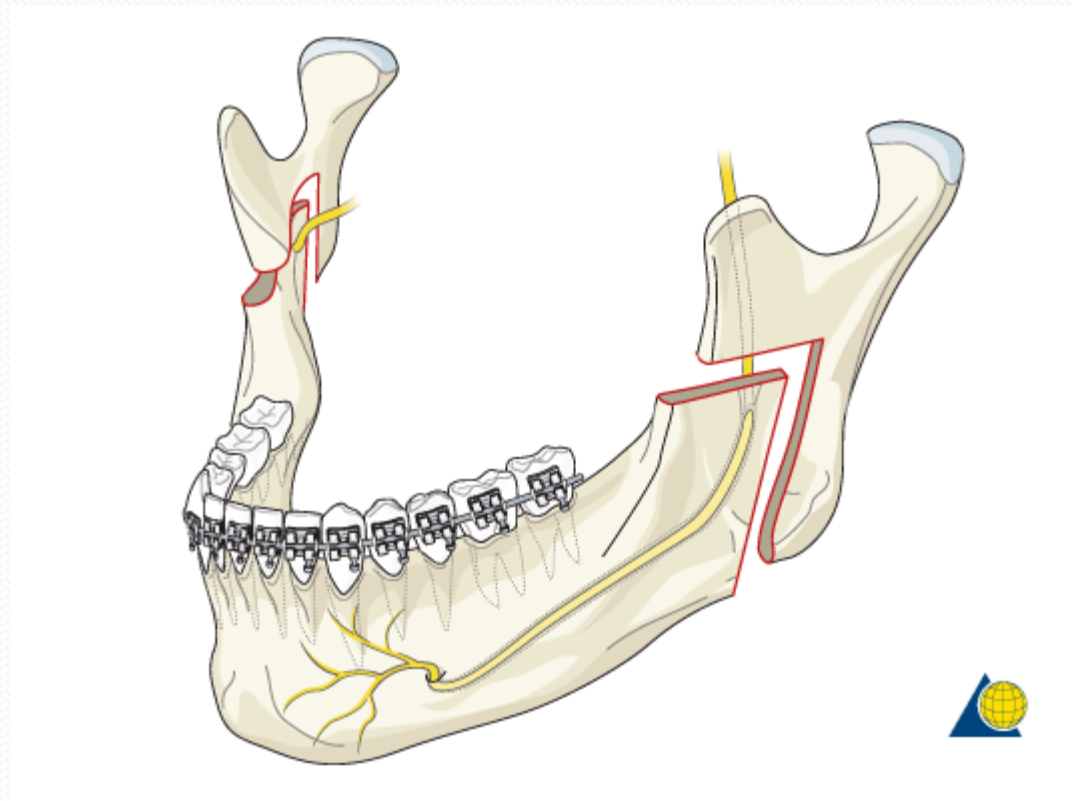
- The C osteotomy for treatment of horizontal mandibular deficiencies.
- The inverted L for the correction of most kinds of mandibular horizontal discrepancies, including anterior open bite.
- Advancements of the distal segment with either technique require bone grafting to ensure adequate bone union.

1) The inverted L osteotomy

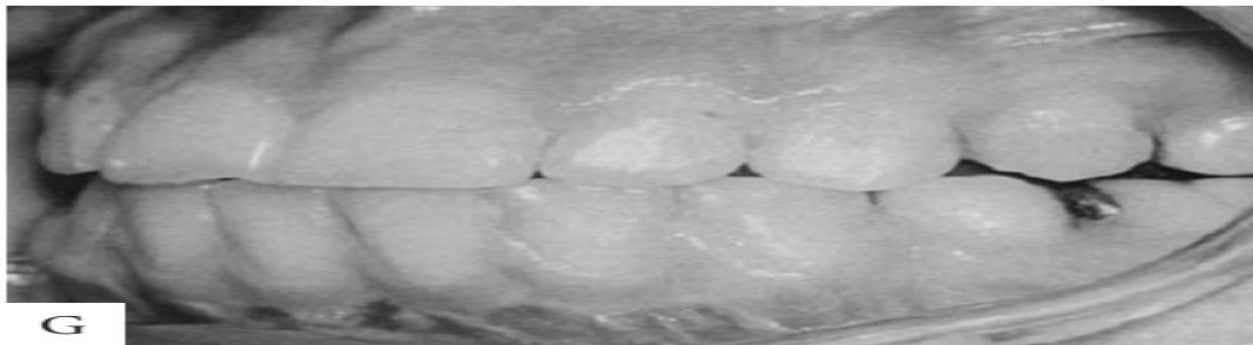
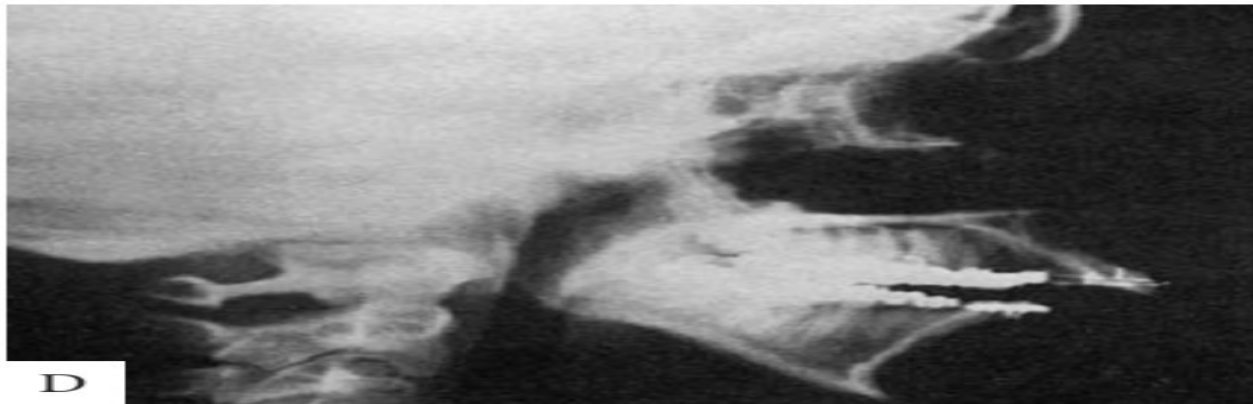
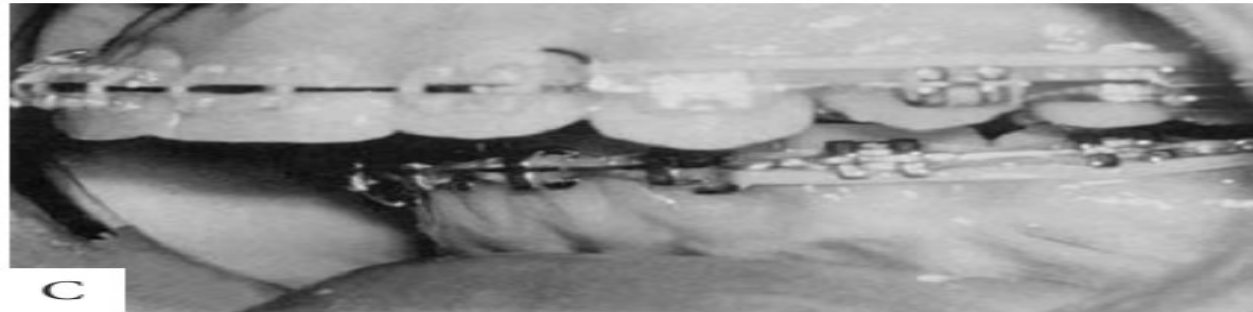
Surgical procedure

- Both are generally approached extraorally.
- The posterior vertical osteotomy is made at least 7 mm in front of the posterior border and extends to a point of the inferior border just in front of the angle.
- The horizontal cut is made above the anticipated position of the inferior alveolar foramen.
- For distal fragment advancement , place the mandibular teeth into the new occlusal position and secure with maxillomandibular dental fixation.
- If the distal segment is set back, then the proximal segment has to be overlapped laterally.
- Rigid internal fixation with metal plates or mesh secured with screws has become more popular.

Inverted L ramus osteotomy



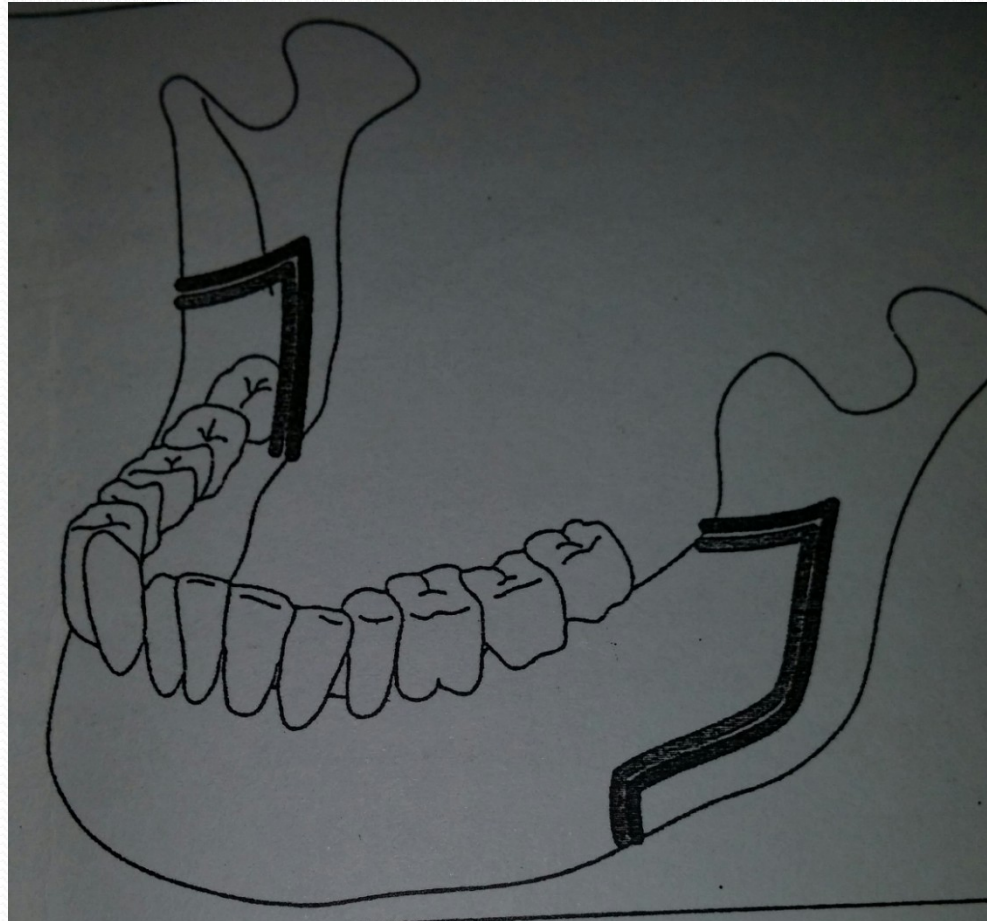
Mandibular horizontal deficiency



2) The C osteotomy

- Vertical L with the addition of a horizontal cut that extended forward from the vertical cut below the inferior alveolar canal.
- In case of advancement bone graft taken from the lateral cortex of the distal segment and transferred back into the gap of the mid-ramus area.
- The use of vitalium mesh with two screws in each fragment is effective, but any rigid plate with screws can be used.

Inverted C ramus osteotomy



Indication

Body osteotomies and osteotomies

- Some unusual mandibular deformities.
- In combination with ramus osteotomies in patients with unusual dento-alveolar configurations combined with mandibular deformities.

Disadvantages

- The integrity of the dental arch has to be disturbed.
- Nerve damage more common.

Advantage

- Because these osteotomies are made in front of the pterygo-masseteric sling, the results are more stable.

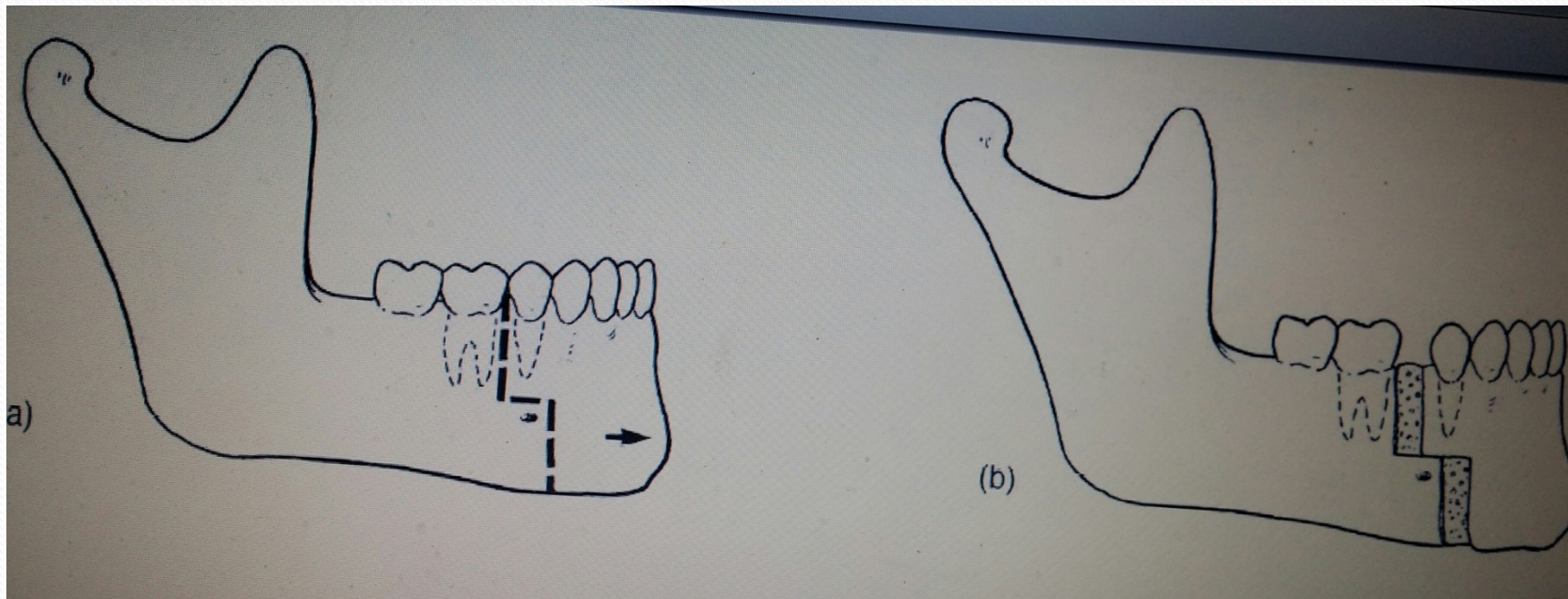
Surgical procedure

Step osteotomy

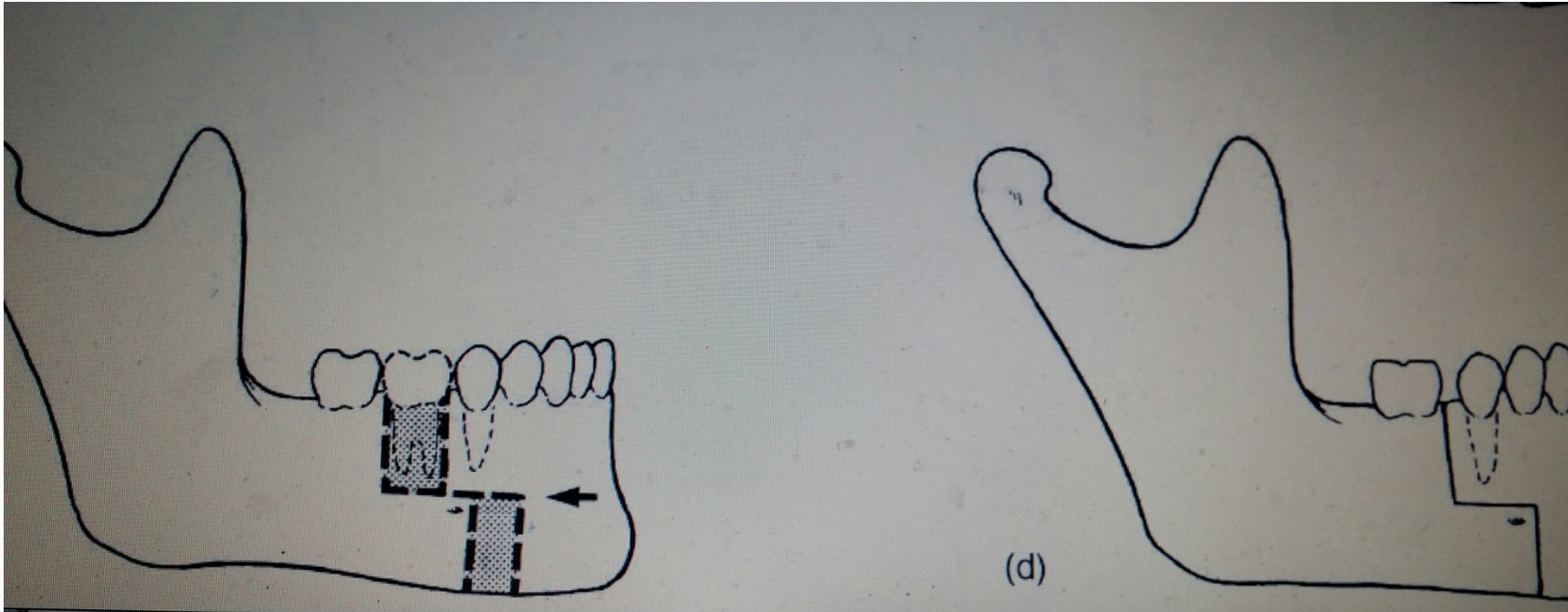
- An incision is made 4 to 5 mm below the level of the attached gingiva, until the cuspid, where it is dropped down 5 mm and extended forward to the midline.
- The horizontal bone cut is made first through the buccal cortex and should run just above the level of the nerve.
- The superior vertical cut is then made and in case of setback the appropriate amount of bone should be removed.
- If the bone cut will be close to a tooth, a margin of at least 2 mm bone should be left attached to the neighbouring teeth.
- Then the inferior vertical cut is made.
- The distal fragment is fixed in the planned position with the aid of acrylic splint and IMF.
- A four-hole mini-plate is placed under the level of the nerve and fixed with four screws and another four-hole plate is fixed across the vertical cut if there is no root tips; otherwise it is advised to fix the acrylic splint to the mandibular dental arch with wires .

a) Design of step osteotomy in the horizontal branch of the mandible

b) in case of advancement bone grafts should be used



- c) Design of step osteotomy in case of set-back osteotomy
- d) step design allows good fit of the segments



Subapical Osteotomies

Anterior Subapical Osteotomy

Indications

- to move the anterior mandibular teeth and alveolus in almost every conceivable direction.

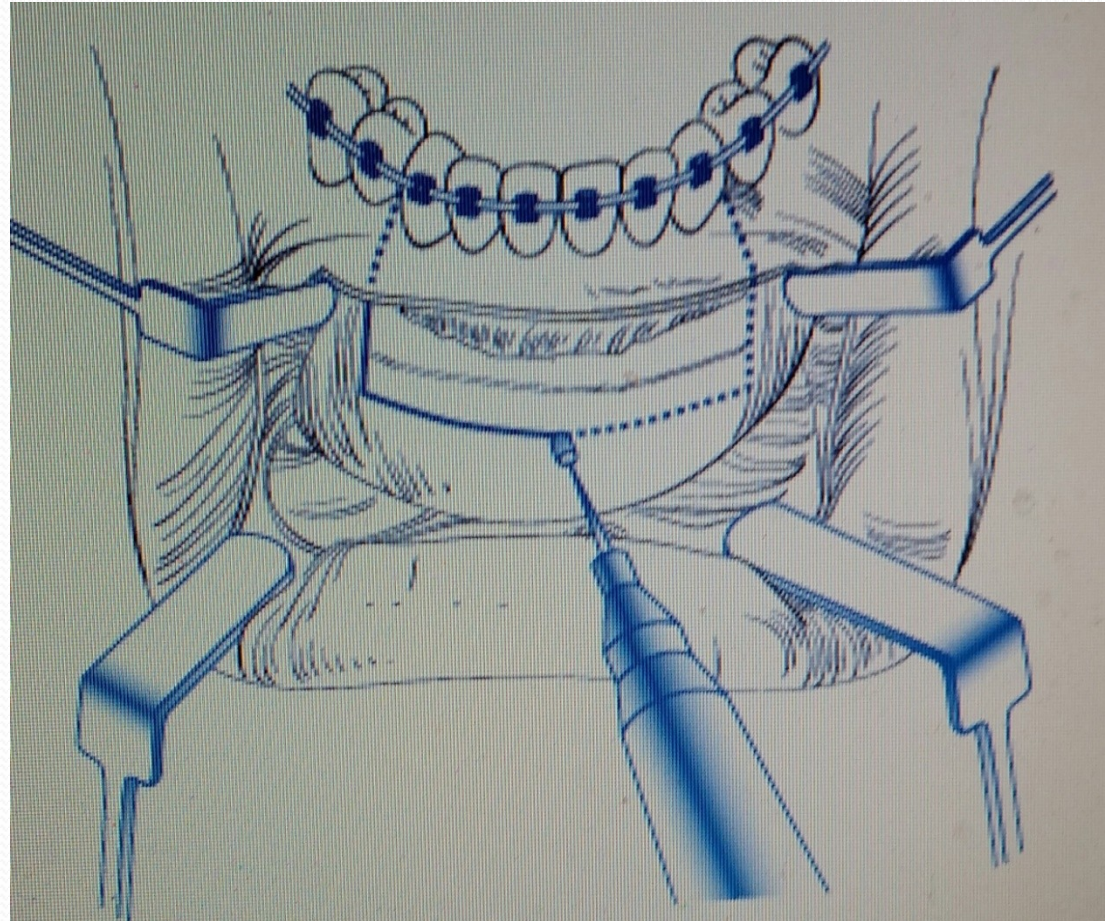
Surgical procedure:

- If necessary, teeth are removed to permit the osteotomies or to provide space for the planned alveolar movement.
- The incision is started about 1 cm behind the planned vertical osteotomy and is carried forward about 4 to 5 mm below the attached tissue until reaching the cuspid, at which time it can be dropped down and carried to the midline.

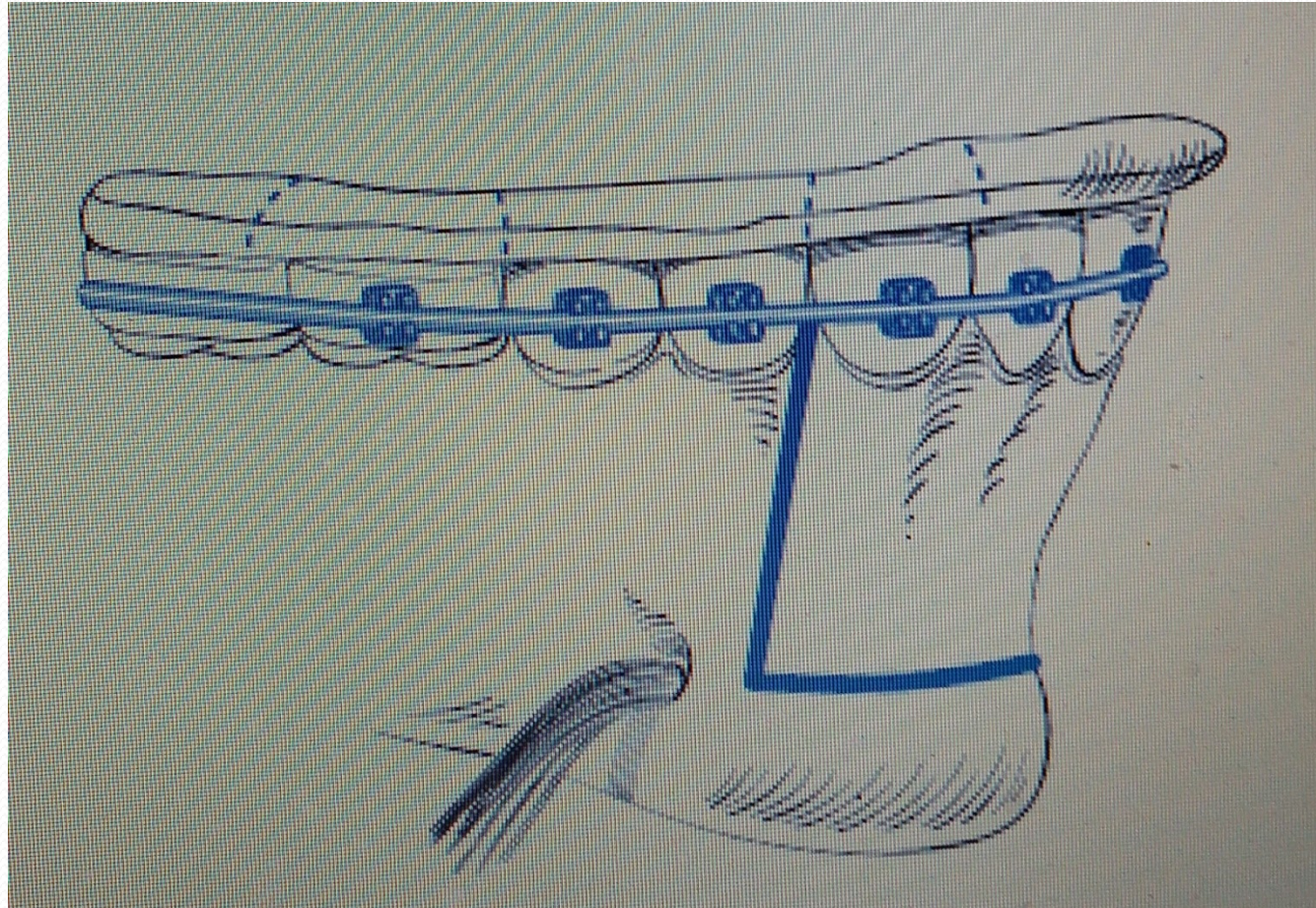
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- The horizontal cut should be at least 5 mm below the teeth apices.
- The vertical osteotomies are made using parallel cuts when the posterior movement of the segment is planned.
- If parallel horizontal cuts are planned to move the anterior segment apically, the superior cut is made first. The inferior cut is then made, and the segment of bone is removed.
- The segment is stabilized by either wiring the splint to the teeth or by circumferential mandibular wires combined with intermaxillary fixation.
- Bone gaps caused by movement of the segment should be grafted.

Anterior subapical osteotomy.



Anterior subapical osteotomy. Occlusal splint



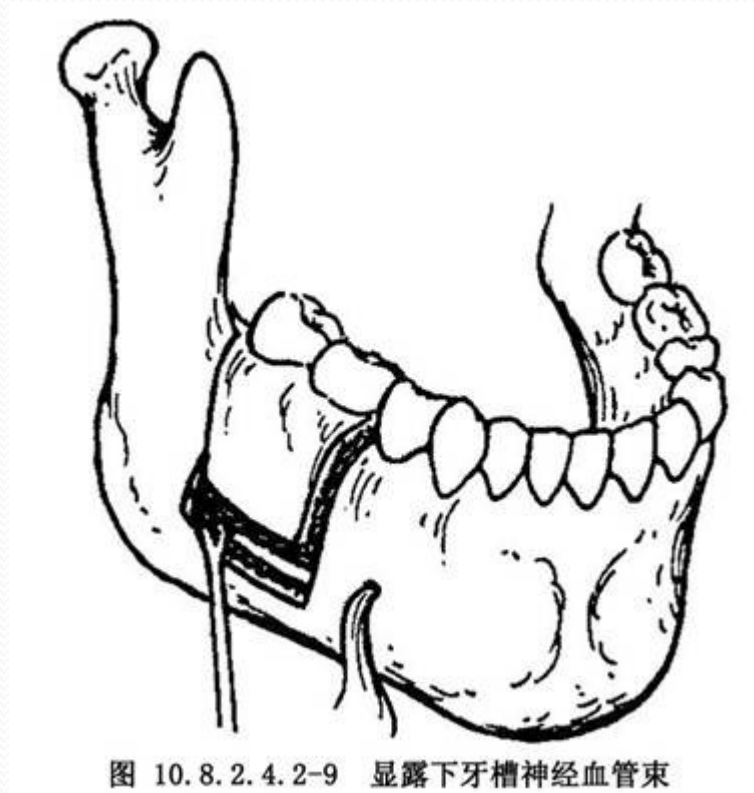
Indications

- correction of **super eruption** of posterior mandibular teeth or **ankylosis** of one or more posterior teeth.
- Abnormal **buccal or lingual position** of these teeth
(when orthodontics is not feasible)

Surgical procedure:

- An incision is started 3 to 4 mm below the attached gingiva, beginning at the anterior border of the vertical ramus, and is carried forward to the cuspid.
- The vertical cuts are made first through both cortices.
- The horizontal cut is carried only through the buccal cortex, and a thick splitting osteotome is used to complete the osteotomy.
- The segment is positioned and stabilized with an acrylic splint and wire.
- Grafting is used if a bone gap remains.

Posterior subapical osteotomy.



Indications

- Mandibular **dental alveolar protrusion or retrusion**.
- Closure of **anterior open bite** when used with a bone graft.

Surgical procedure:

- An incision is starts on the external oblique ridge at the base of the vertical ramus, and extends forward 4 to 5 mm below the attached gingiva.
- The vertical cut posterior to the last molar is made first, and the horizontal cut placed between the dental apices and the inferior alveolar nerve, or the nerve needs to be exteriorized or the cut placed below the nerve.
- The mobile segment is fixed to the maxilla with an acrylic occlusal splint and intermaxillary fixation.
- As with correction of an anterior open bite any bone gaps created are filled with graft material.

Cont.

Total Subapical Mandibular Osteotomies

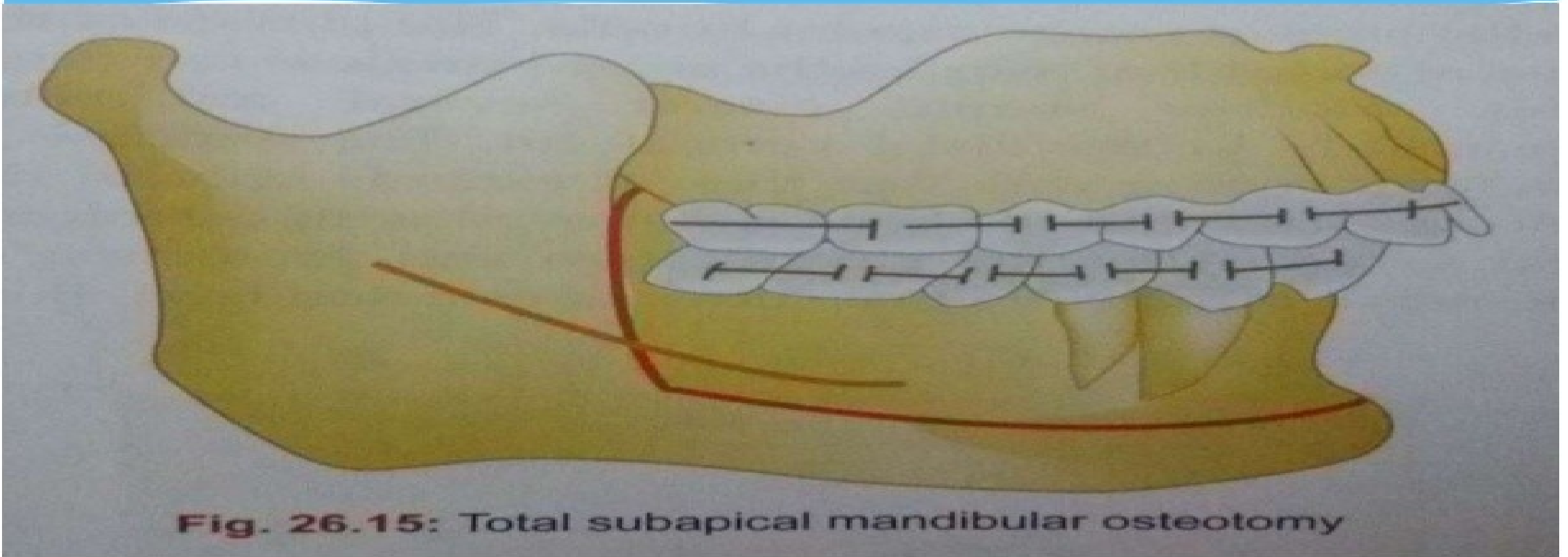


Fig. 26.15: Total subapical mandibular osteotomy

Genioplasty

Surgical techniques used to alter the size and morphology of the bony chin and the surrounding soft tissues.

Chin deformities:

- **Functional deformities**
 - Chin ptosis
 - Lower lip incompetence
- **Esthetic deformities**
 - Macrogenia (large chin)
 - vertical
 - horizontal (antero-posterior)
 - combination of vertical and horizontal
 - Microgenia (small chin)
 - vertical
 - horizontal
 - combination of vertical and horizontal
 - Asymmetry

Genioplasty types

Functional genioplasty

Used to **correct abnormal mentalis muscle activity** at early age (just after eruption of mandibular canines)

Indications:

- **Vertical chin excess** requiring vertical reduction genioplasty.
- **Horizontal chin deficiency** requiring advancement genioplasty.
- **Asymmetric genioplasty:-** the vertical shift of the genial segment.

Horizontal Osteotomy of the Symphysis

Indications

- primarily used only for esthetic reasons.

Technique

- Incision is made on the labial side of the vestibule at about 1cm above its depth and extends posteriorly to the first bicuspid.
- The direction of the bone cut depends on the vertical dimensions.
- For advancement with maintenance of the vertical dimensions , the cut should be as horizontal as possible.
- If some reduction of vertical height is wanted , the bone cut may be inclined.
- When substantial vertical reduction is required , a second bone cut is made parallel to the first one

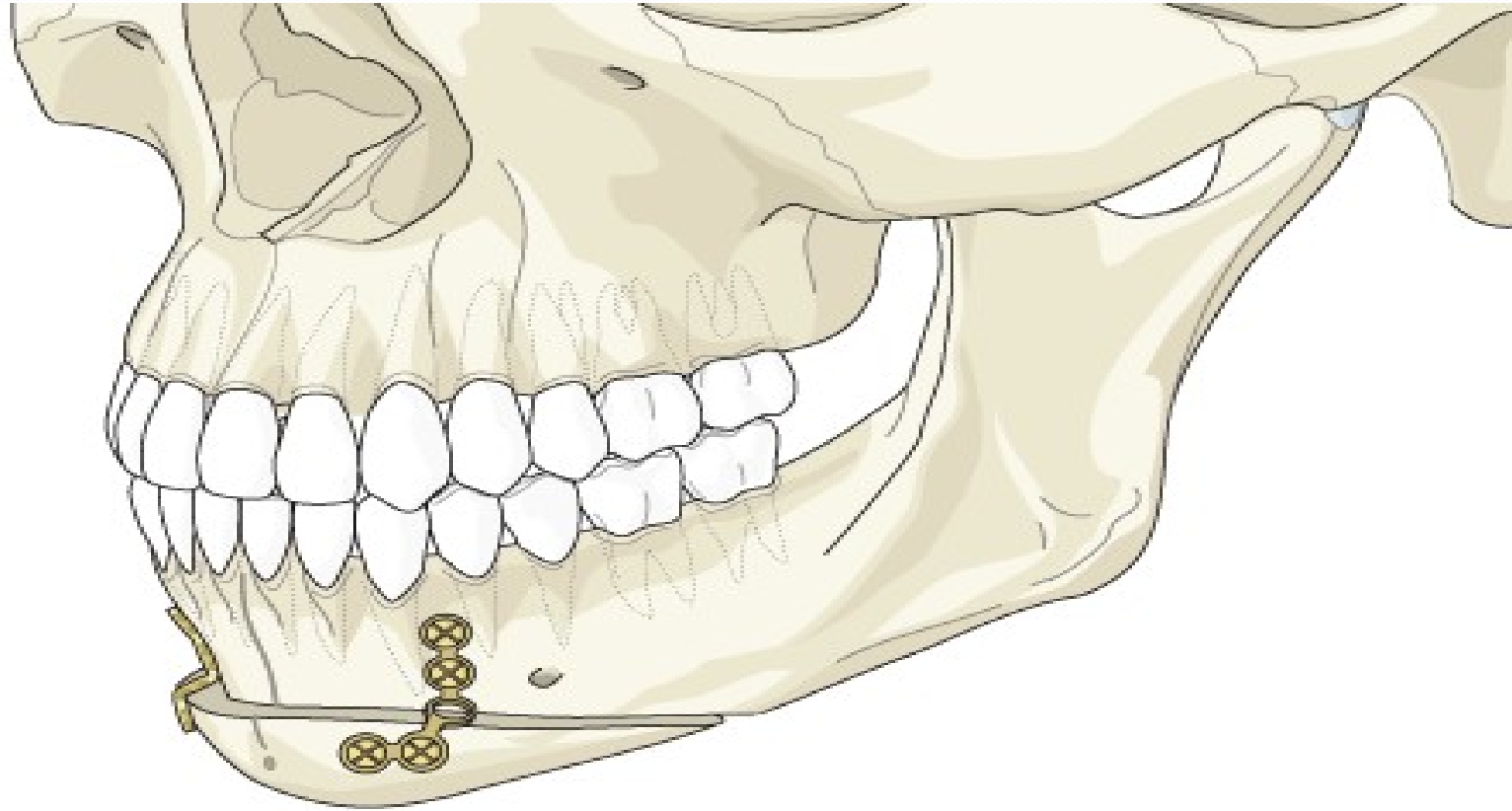
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- In cases of antero-posterior reduction, four vertical cuts 1 cm each connected by a horizontal cut and by removing bone between two vertical cuts on each side the reduction is achieved (box technique).
- A midline osteotomy is necessary to allow for the fragments to be widened.
- The stabilization of the segment can be made with cortical wires, circummandibular wires, or plates and screws.

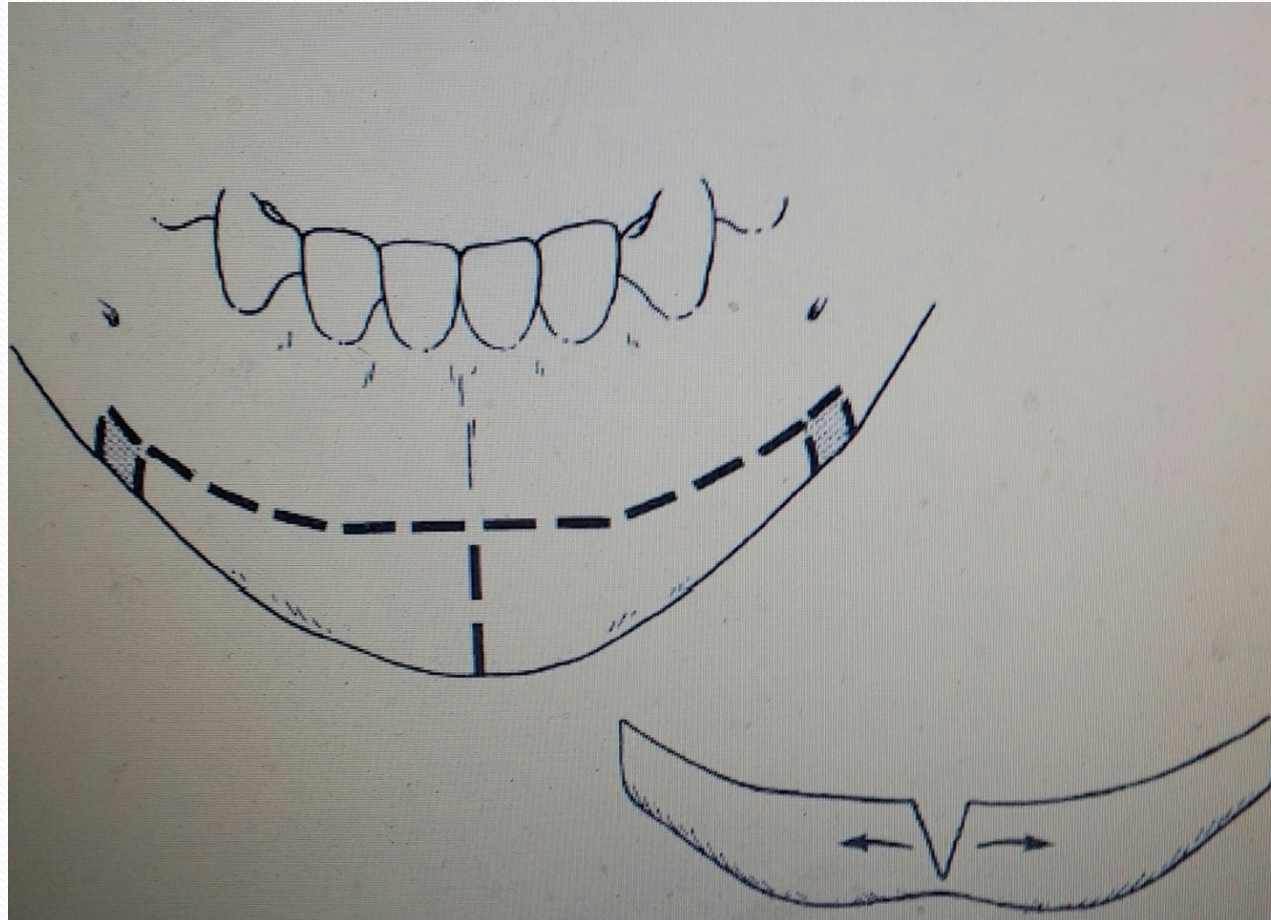
Direction of the horizontal cut if lower facial height is to be reduced



Direction of the horizontal cut if lower facial height is to be maintained



Design of box osteotomy to reduce chin prominence



Maxillary Orthognathic Surgery

Classification

- **Segmental maxillary surgery**
 - single-tooth osteotomy
 - corticotomy
 - anterior segmental osteotomy:
 - Wassmund
 - Wunderer
 - Epker and Wolford
 - posterior segmental osteotomy:
 - Schuchardt
 - Kufner
 - horseshoe osteotomy
 - Wolford and Epker

• **Total maxillary surgery**

- A) Le Fort-I osteotomy
 - classic downfracture
 - surgically assisted maxillary expansion
 - quadrangular
- B) Le Fort-II osteotomy
 - anterior Le Fort **II** osteotomy
 - pyramidal Le Fort **II** osteotomy
 - quadrangular Le Fort **II** osteotomy
- C) Le Fort **III** osteotomy
 - Gillies
 - Tessier
- D) Other midfacial osteotomies
 - zygomatic osteotomies
 - malar-maxillary osteotomy

Segmental maxillary osteotomy

Indication

- Correction of dentofacial deformities , where only a **portion of the maxillary dental arch requires repositioning** for functional and esthetic reasons.
- Correction of **tooth malposition** , **dental ankylosis** or closure of **diastema**(single tooth osteotomy).
- **widening the maxilla and adjusting the angulations of the posterior maxillary segments.**

Single tooth dento-osseous osteotomies

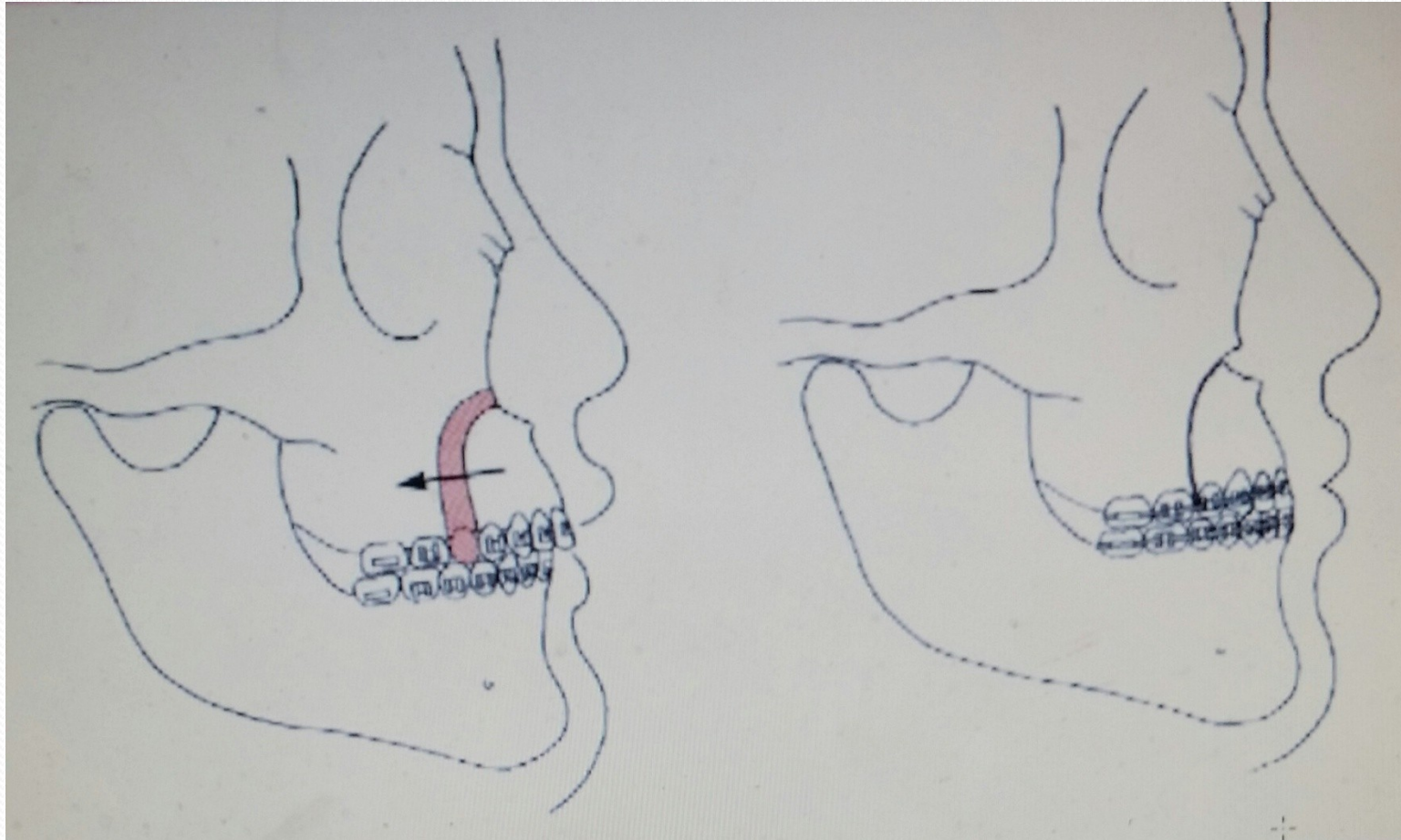
- There must be **sufficient space** between adjacent tooth roots.
- There should be **adequate space in the dental arch** to reposition osteotomised segment into proper occlusion .
- A **horizontal vestibular incision**(one tooth distal to those being mobilized) **or multiple vertical incisions.**
- **Vertical interdental osteotomies** with fine fissured bur, and the alveolar crest bone sectioned with a fine osteotome.

Anterior maxillary osteotomy

Indication

- Correction of **dentoalveolar protrusion of the anterior maxilla.**
- Correction of **bimaxillary protrusion** in conjunction with mandibular subapical osteotomy.
- Correction of **anterior open bite.**
- In patients with a **severe curve of Spee**, coupled with mandibular advancement and anterior mandibular segmental surgery .
- The **three major techniques** involve the **use of one of three vascular pedicles: labial , palatal ,and a combination** of these with vertical incisions in both.

Anterior maxillary osteotomy



Le Fort I osteotomy

Indication

- Correction of maxillary hyperplasia and hypoplasia in all three dimensions.
- Correction of maxillary asymmetries.

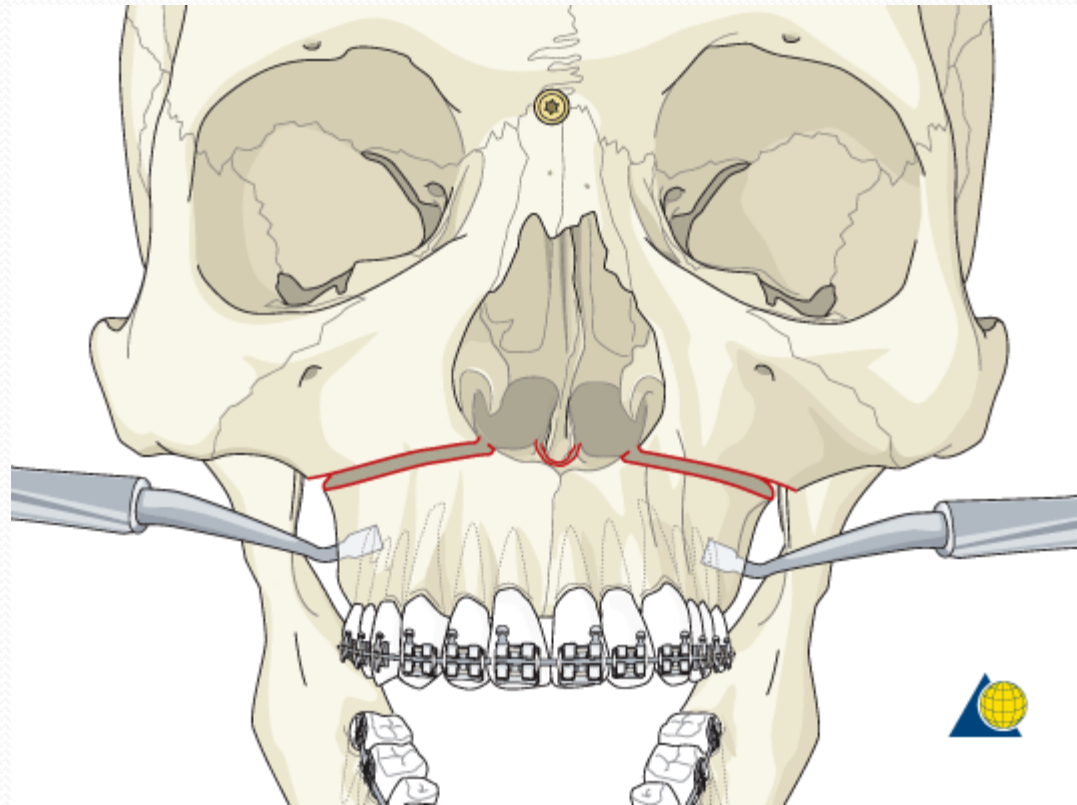
Technique

- Incision in the buccal vestibule approximately 1 cm above the junction of fixed and mobile mucosa and ends at the level of the first molar.
- Locate a Le Fort I level osteotomy at least 5 mm above the apices of the teeth:
 - the step osteotomy permits horizontal maxillary movements without altering vertical dimensions.
 - Anterior cuts: 4-5 mm above the canine apices.
 - Vertical cuts: zygomatic buttress region.
 - Posterior cuts: 4-5 mm above the molar teeth.

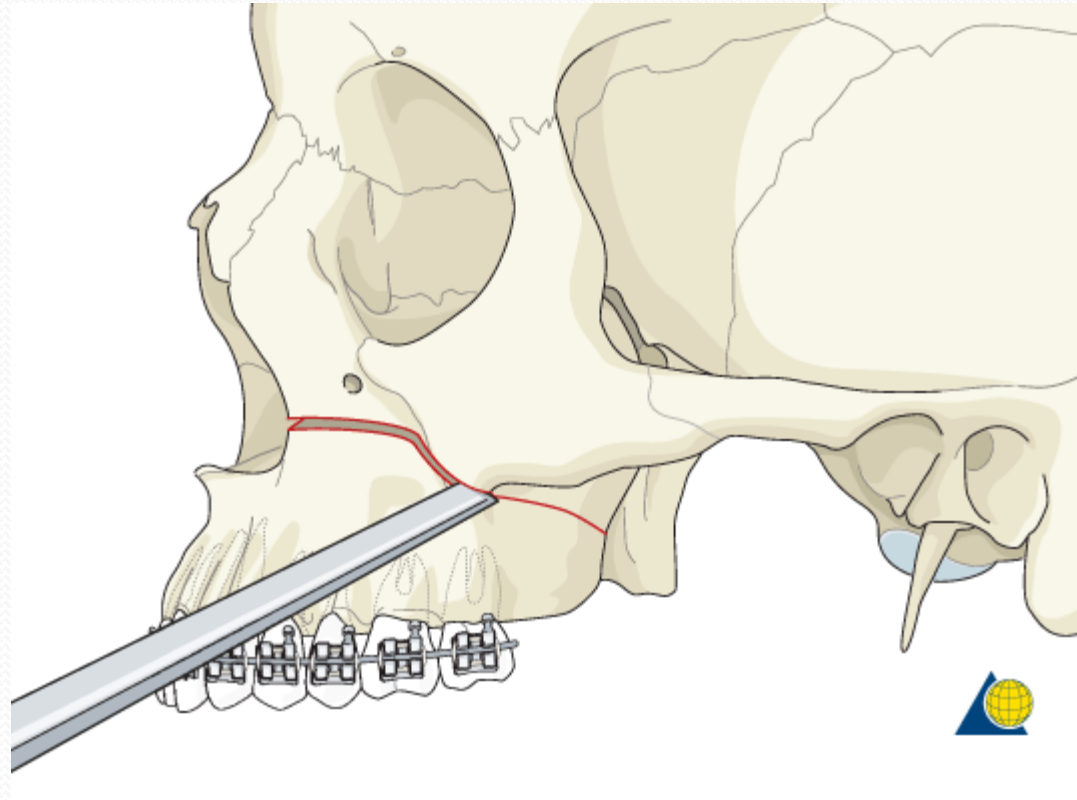
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- Perform an **osteotomy of the nasal walls and nasal septum.**
- **Separate the pterygo-maxillary junction**
complication of unfavorable dysjunction in this region includes:
 - a) excessive bleeding
 - b) cranial nerve injury
 - c) carotid artery injury

Outline of the Le Fort I osteotomy.



Outline of the Le Fort I osteotomy.



Maxillary advancement

Indications

- Antero-posterior maxillary deficiency(maxillary hypoplasia)
 - without cleft palate
 - with cleft palate
- Esthetics
 - When the naso-labial angle is reduced
 - when the nasal bridge can not be seen in profile because the eye protrudes beyond it
 - maxillary incisal display is inadequate
 - prominent everted lower lip
 - nasal balance (nasal tip , alar base width ,naso-labial fold)

- Occlusion

- class III malocclusion
- severe maxillary dental crowding
- constricted maxillary dental arch with posterior crossbite

- Cephalometric features

- maxillary hypoplasia and a decreased sella-nasion-point A angle
- point A is well behind the McNamara perpendicular line

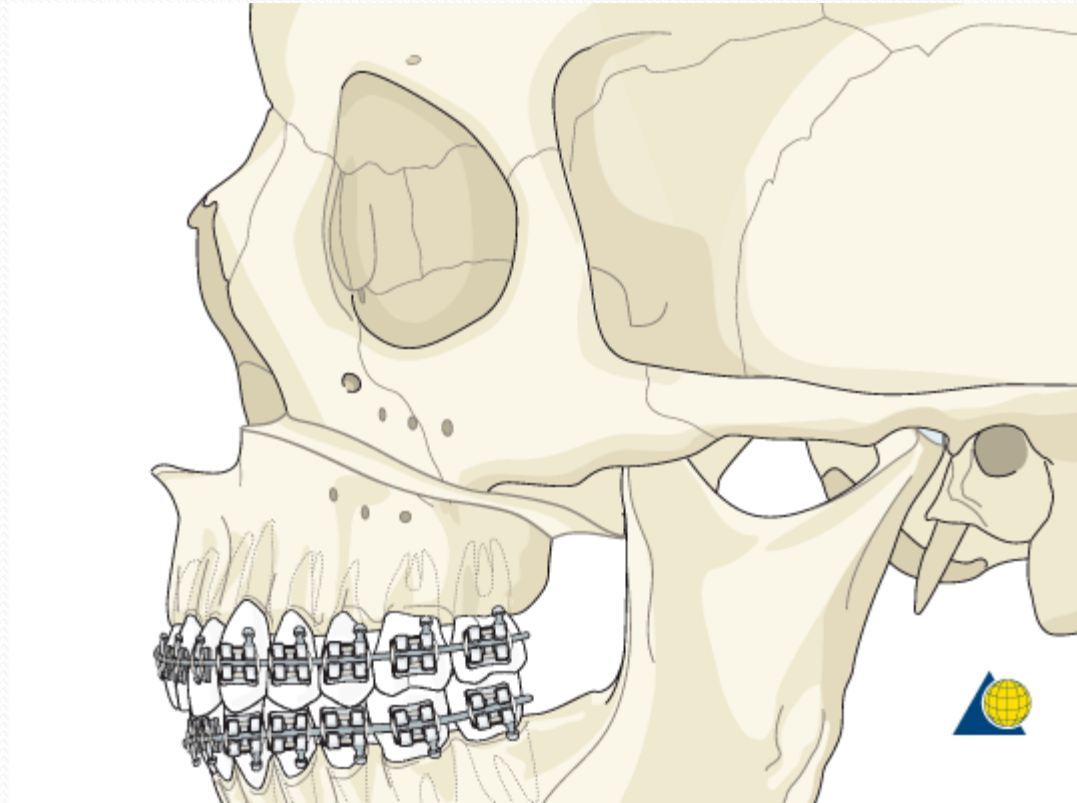
- Velopharyngeal function

- of concern in cleft patients with short palates, because maxillary advancement may accentuate velopharyngeal incompetence.
- maxillo-mandibular surgery should be considered with minimal amount of maxillary advancement.

Surgical procedures

- Place interpositional **bone grafts** within osteotomy gaps :
 - for maxillary **advancements greater than 5 mm**
 - **in difficult cleft cases**
- Use **bone grafts**:
 - **to accelerate bone healing** between segments
 - **to increase stability and minimize relapse**
- Completely mobilize the maxilla to be passively placed in the desired anterior position.

Maxillary advancement



Maxillary expansion

Indications:

- **Transverse maxillary deficiency** (bilateral or unilateral) either isolated or with a class III malocclusion
- **Narrow maxilla with dental crowding**

Treatment planning

- **Children** can be successfully treated with **orthopedic rapid maxillary expansion**
- **Adults :**
 - For **less than 5 mm** of transverse deficiency , **orthopedic expansion** of the maxilla is sufficient
 - For **greater than 5 mm** of transverse deficiency or **buccally inclined posterior teeth** , a **combined surgical-orthodontic expansion technique** is used.

Combined surgical-orthodontic expansion

There are three surgical options:

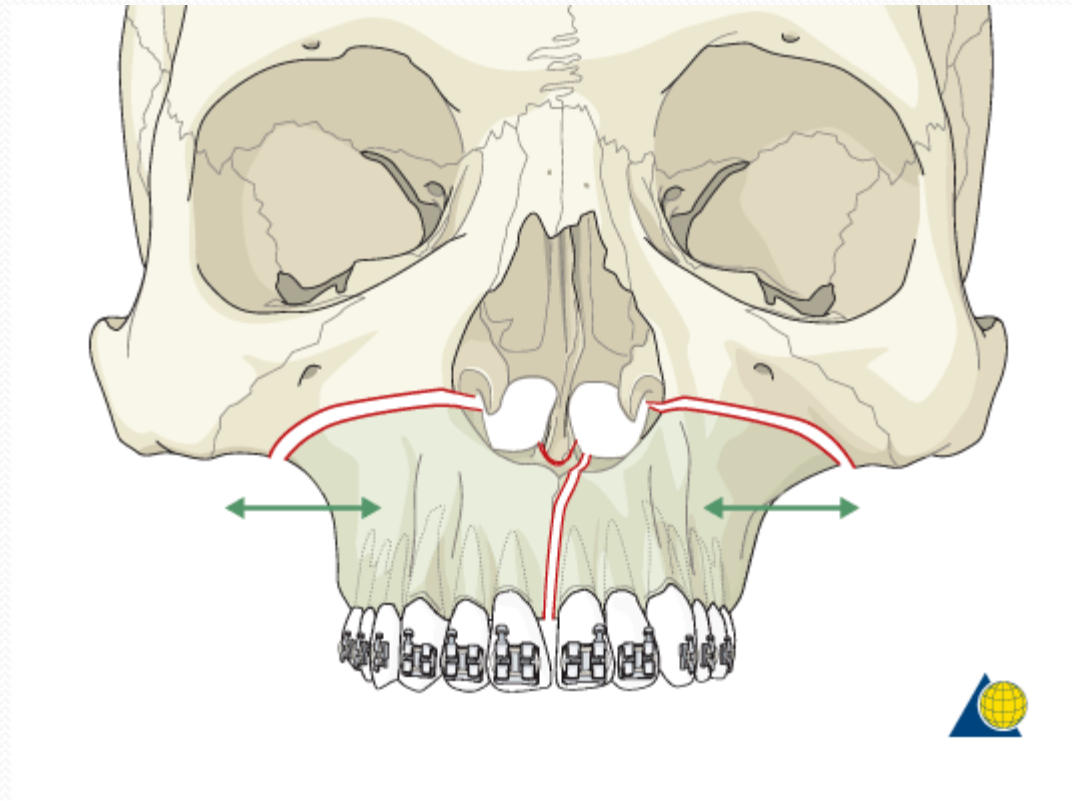
- 1) Segmental posterior osteotomy (for unilateral deformities).
- 2) Le Fort I osteotomy with midline palatal split or horseshoe osteotomy.
- 3) Surgically assisted rapid maxillary expansion:
 - uses a combination of surgical relief of maxillary bony abutments and palatal expansion devices to rapidly open the midpalatine suture.
 - incisions are made from the piriform rims to zygomatic buttresses
 - complete a Le Fort I osteotomy without the downfracture.
 - make a midline palatal split between the maxillary central incisors.
 - intra-operatively , activate an orthopedic expansion device bonded to the posterior maxillary teeth.

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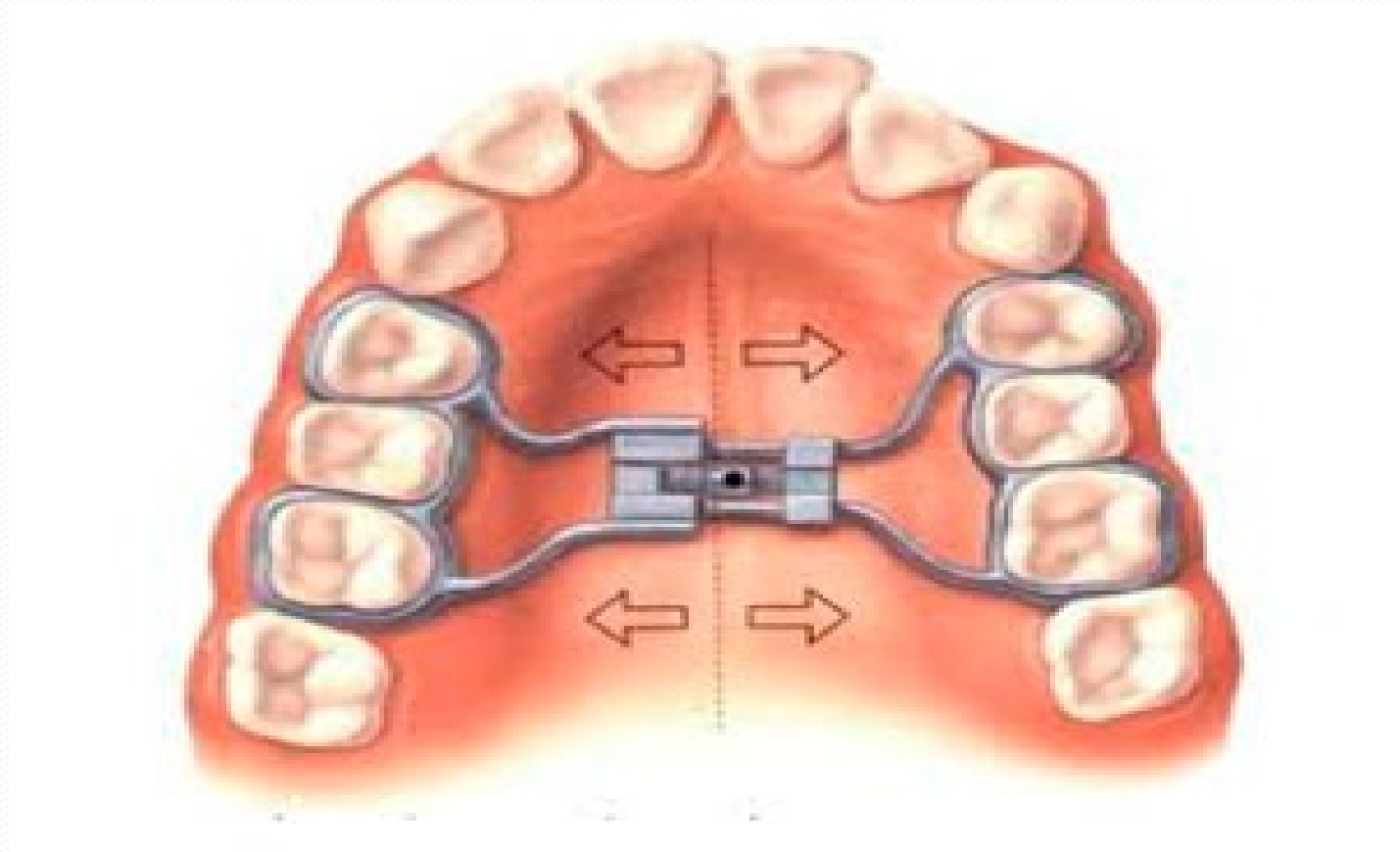
Indications

- To increase the intercanine width and to flatten the anterior segment for a normal elliptical arch morphology.
- Ectopic eruption of one or two posterior teeth.
- Arch length deficiency

Le Fort I osteotomy with midline palatal split or horseshoe osteotomy.



Surgically assisted rapid palatal expansion. Expansion device



Two-jaw(maxillo-mandibular) surgery

Indications:

Facial asymmetries

- Anterior open bite.

- Combined horizontal and vertical deformities and/or transverse discrepancies.

- Severe single-jaw deformities with significant overjet discrepancies(greater than 10 mm).

Surgical procedures and sequence

Complete a mandibular osteotomy without the split.

- Complete a Le Fort I maxillary osteotomy and downfracture.

- Fully mobilize the maxilla and remove bony interferences.

- Place the maxilla in the desired position with surgical splint and apply IMF using the mandible as a guide.

Fix the maxilla in position with the plates and screws.

- Release IMF and remove the intermediate splint.

- Move the mandible into the desired position using surgical wafer and apply IMF.
- Apply rigid fixation to the mandible.
- Release IMF .
- Check the final occlusion before suturing.
- Apply light maxillomandibular elastics.

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